

PHSX 661/ PHSX761 Spring 2023

Particle Physics I

Tuesday/Thursday 11:00 - 12:15, Malott 2074

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Office Hours: After class, by appointment and TBD

Zoom room: <https://kansas.zoom.us/j/3879191996>

Course Synopsis: The course is designed to give an introduction to particle physics and its phenomenology understandable to students with a reasonable grasp of quantum mechanics. Particle physics has at its heart a coherent picture of the fundamental particles and their interactions as encompassed in the Standard Model. The course is intended as a survey course that is well suited to introducing this modern field to advanced undergraduates and graduate students.

- Fundamental particles: the quarks and leptons and gauge bosons
- Basic concepts: relativistic kinematics, Lorentz invariants, decay rates, cross-sections etc
- Particle detection and particle detector principles
- Accelerators and Particle Physics Experiments
- Particle interactions (strong, electromagnetic, weak)
- Overview of Feynman diagrams at the phenomenological level
- Introduction to particle interactions in matter simulations and general-purpose particle physics event generators
- Invariance principles, symmetries, and conservation laws
- Dirac Equation/spinors/helicity/chirality
- Unification of electromagnetic and weak interactions
- Hadrons and their decays
- Colliders
- More advanced topics including neutrino oscillations, polarization observables, CP violation, Higgs physics, particle astrophysics, ideas for physics beyond the Standard Model.

Prerequisite(s): PHSX511

Textbook(s): It is strongly suggested that you acquire access to at least one textbook to aid your study. The following two books are highly recommended for the course:

Modern Particle Physics, Thomson (2013)

Introduction to Elementary Particles, Griffiths (2008) (2nd edition)

I will be using both of them. Thomson is to be recommended for its more modern approach and coverage of more advanced phenomenology (theory and experiment), and is my **main** recommendation. Griffiths excels in the pedagogic presentation of the more basic material. It is more limited in scope and has a theoretical emphasis. I will also use a number of other books for my class prep. including, but not limited to:

Concepts of Elementary Particle Physics, Peskin (2019)

Elementary Particle Physics: An Intuitive Introduction, Larkoski (2019)

Elementary Particle Physics in a Nutshell, Tully (2011)

Particle Physics in the LHC Era, Barr, Devenish, Walczak, Weidberg (2016)

Introduction to High Energy Physics, Perkins (2000)

Homework: Will be assigned biweekly (expect 6 assignments). Solutions will be posted promptly so homework must be received on time to get credit.

Grading:

Homework	60%
Term project (simulation / data analysis)	8%
Term paper & presentation	12%
Peer review refereeing	5%
Take-home final	15%

Exams: Take-home final: distribute by May 4, due date is May 12.

Term project: Develop in consultation with the instructor a data analysis style project based around simulations carried out with either GEANT4 or an event generator like WHIZARD, PYTHIA8, or MADGRAPH. GEANT4 is a complete toolkit for all areas of detector simulation that encompasses models for diverse interactions of particles with matter over a wide energy range. WHIZARD, PYTHIA8, and MADGRAPH are general purpose programs for the generation of high-energy physics events especially for colliders.

Term paper: Write a short review paper on a topic of interest in modern particle physics that should be interesting to the whole class. This should be presented in a short presentation (10 mins). Ideal topics tend to be relatively recent results of significance that are not too close to your existing research or experience.

Peer review refereeing: For the term papers you will be asked to review and give courteous constructive comments on the term papers and term projects to your fellow students. The presentations should also be useful for this.

V1.0 Jan. 17. Updates if any will be announced in class or by E-mail